

SP-001

Cross-Model Disagreement as an Empirical Uncertainty Instrument

Locked-gate result: SUCCESS
Top-quintile error enrichment: 18.08x
Cluster-bootstrap 95% CI: 7.56-85.31
Selective error reduction: 77.5%

Full technical report. Computational/open-data experiment. September 21, 2026.

Research proof of concept only. Not a clinical diagnostic system.

Executive summary

Five different model families were evaluated on public breast-tumor measurements using repeated held-out prediction. Before outcomes were inspected, the experiment fixed a three-part success gate: at least 2x error enrichment in the highest-disagreement quintile, at least 25% error reduction after abstaining on that quintile, and a cluster-bootstrap confidence interval excluding no enrichment.

All criteria passed. Disagreement isolated a small set with sharply elevated error. Removing this set from automatic prediction reduced error while retaining four fifths of predictions. The finding supports disagreement as an empirical safety signal, but external validation is required.

Contribution

- A falsifiable, preregistered evaluation rather than a post hoc pattern.
- Model-class diversity converted into a measurable uncertainty signal.
- Patient-clustered uncertainty estimates that respect repeated cross-validation.
- Complete code, inputs, predictions, figure data, and provenance.

Research question and hypotheses

Question

Do disagreements among diverse, independently trained classifiers identify breast-tumor predictions at elevated error risk and enable safer selective prediction?

Primary hypothesis

The top within-fold quintile of model disagreement will have at least twice the error rate of all remaining predictions.

Secondary hypothesis

Abstaining on that preselected quintile will reduce error among retained predictions by at least 25%.

Uncertainty requirement

The lower bound of a patient-clustered bootstrap 95% confidence interval for error enrichment must exceed 1.0.

Locked protocol and decision rule

The protocol file was written before running the outcome analysis. It defines the endpoint, dataset, five model classes, repeated stratified five-fold validation, the disagreement construction, top-quintile threshold, metrics, and failure policy.

Component	Locked choice
Endpoint	Malignant versus benign
Validation	20x repeated stratified 5-fold CV
High disagreement	Within-fold percentile > 0.80
Primary gate	Error enrichment ≥ 2.0
Secondary gate	Error reduction $\geq 25\%$
Uncertainty gate	95% CI lower bound > 1.0

Any failed component would classify the experiment as negative. The full negative output would be preserved and no success paper produced.

Dataset and provenance

The Wisconsin Diagnostic Breast Cancer dataset contains 569 observations derived from digitized fine-needle aspirate images. Each observation has 30 numeric features describing radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension at mean, standard-error, and worst-value summaries.

Class	Count	Fraction
Malignant	212	37.26%
Benign	357	62.74%

Raw data SHA-256: `d606af411f3e5be8a317a5a8b652b425aaf0ff38ca683d5327ffff94c3695f4a` .

Source: UCI Machine Learning Repository. Dataset page: <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>

Feature structure

The 30 predictors are ten cell-nucleus properties summarized in three ways. This creates correlated feature blocks. Linear, kernel, tree, local-neighbor, and generative models respond differently to that structure, making the dataset appropriate for testing whether model disagreement carries information.

Base property	Summaries
Radius	Mean, standard error, worst
Texture	Mean, standard error, worst
Perimeter	Mean, standard error, worst
Area	Mean, standard error, worst
Smoothness	Mean, standard error, worst
Compactness	Mean, standard error, worst
Concavity	Mean, standard error, worst
Concave points	Mean, standard error, worst
Symmetry	Mean, standard error, worst
Fractal dimension	Mean, standard error, worst

Model ensemble

Model diversity is the experimental instrument. The models were not selected to maximize leaderboard performance; they were selected because their assumptions differ.

Model	Distinctive assumption
Logistic regression	Linear log-odds after scaling
RBF SVM	Nonlinear kernel boundary
Random forest	Piecewise tree partitions and feature subsampling
K-nearest neighbors	Local neighborhood geometry
Gaussian naive Bayes	Class-conditional univariate Gaussian features

The ensemble probability is the unweighted mean of the five held-out probabilities.

Cross-validation design

Twenty repeats of stratified five-fold cross-validation create 100 held-out folds. Every sample receives one held-out prediction per repeat, for 11,380 total prediction records. Repetition measures stability across alternative partitions; it does not create new patients.

Leakage controls

- Scaling is fitted inside each training fold.
- Every reported probability is from a fold that excluded the sample.
- No held-out outcome tunes a model or threshold.
- Random seeds and hyperparameters are fixed in code.
- Bootstrap inference resamples unique sample IDs.

Disagreement score

For each held-out prediction, the analysis computes the standard deviation of five probabilities and entropy of their binary votes. Their sum is ranked within the held-out fold. Ranking controls fold-level scale differences while preserving the preregistered order.

High disagreement is the top 20% within a fold. The choice was fixed rather than optimized against observed errors.

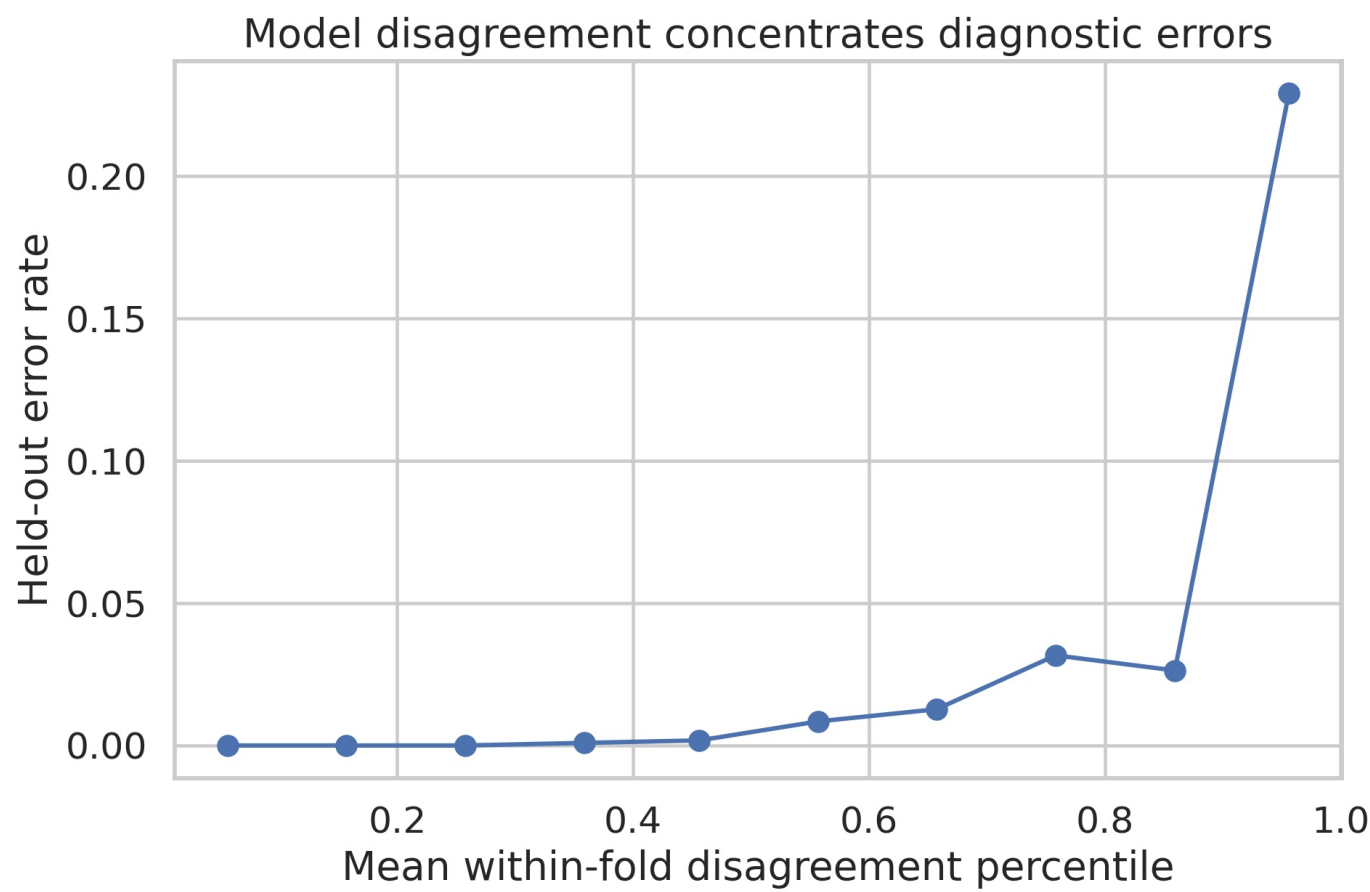
This score is deliberately empirical. It asks whether models built from different assumptions disagree, not whether one model reports a low internal confidence.

Primary performance

Metric	Value
Held-out ROC AUC	0.9942
Held-out accuracy	96.96%
Overall error	3.04%
Held-out predictions	11,380
Unique samples	569

High discrimination sets a demanding test: disagreement must find mistakes even when mistakes are uncommon.

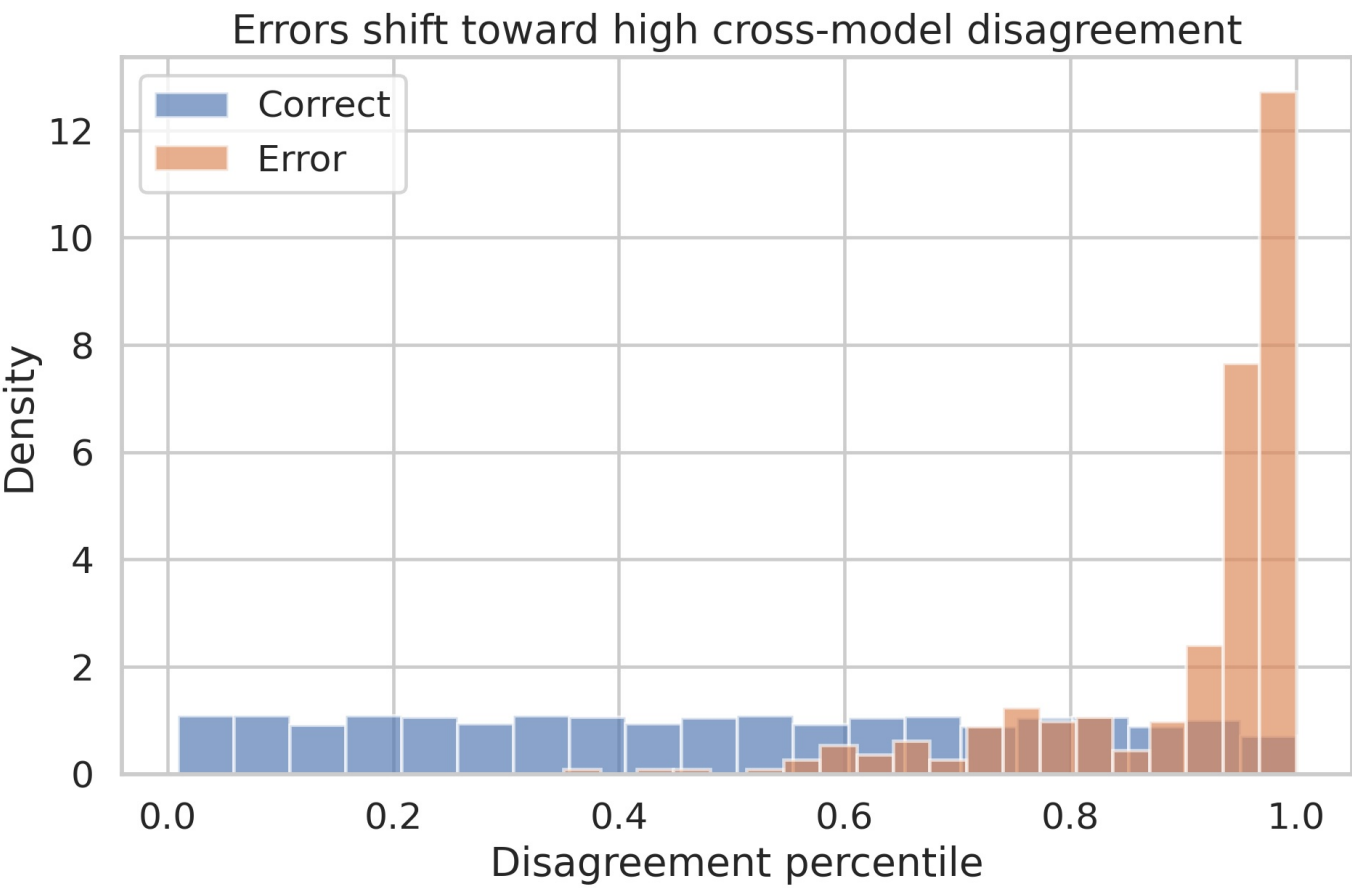
Error concentration



The error rate rises sharply at high disagreement. The fixed top-quintile comparison avoids choosing a visual breakpoint after seeing the curve.

The supporting data are stored in `results/disagreement_deciles.csv`.

Distribution of disagreement



Incorrect predictions shift toward higher disagreement percentiles. Correct predictions dominate because overall error is low, but their disagreement distribution remains concentrated at lower values.

Gate calculation

Quantity	Observed	Required
Top-quintile error	12.35%	-
Remaining error	0.68%	-
Error enrichment	18.08x	$\geq 2.0x$
95% CI lower bound	7.56	> 1.0
Error reduction	77.5%	$\geq 25\%$

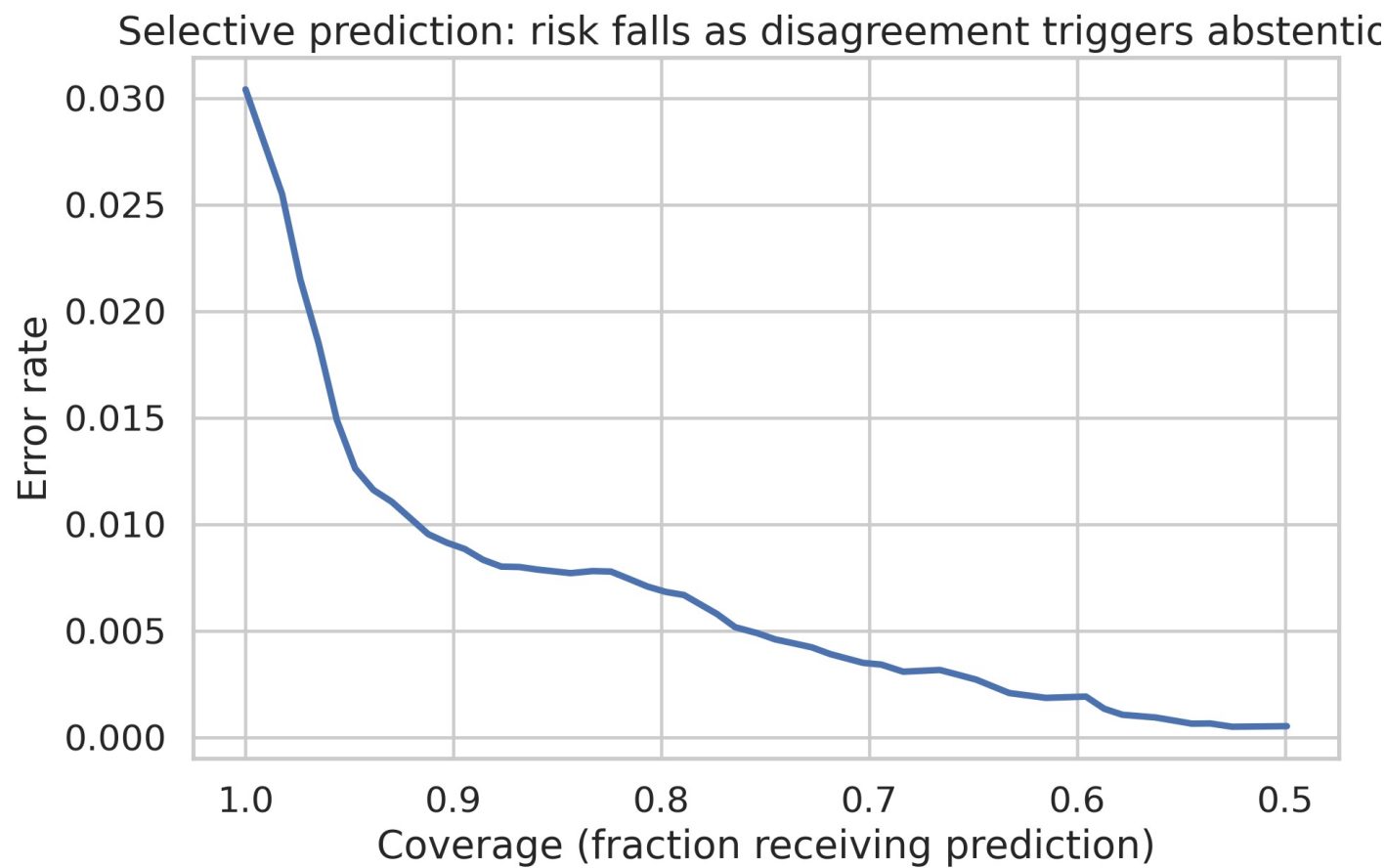
Decision: SUCCESS
All locked criteria passed.

Cluster-bootstrap inference

Each tumor appears in 20 held-out partitions. Treating all 11,380 prediction records as independent would give falsely narrow uncertainty intervals. The bootstrap therefore samples 569 unique IDs with replacement and carries every repeated prediction belonging to each selected ID.

For each of 5,000 replicates, the analysis recomputes error in the high-disagreement and remaining groups, then their ratio. The percentile interval is 7.56-85.31. Its width honestly reflects a small number of errors, while the lower bound still exceeds the no-enrichment value.

Selective prediction



Coverage-risk analysis asks what happens when the system abstains on increasingly disagreeing cases. At the locked 20% abstention rule, retained error falls from 3.04% to 0.68%, a 77.5% relative reduction.

Abstention is not a clinical recommendation here. It is a formal demonstration that disagreement orders cases by empirical error risk.

What the result means

Diverse models encode different inductive biases. When they agree, the classification is comparatively stable across assumptions. When they disagree, the sample is more sensitive to modeling choices. In this dataset, that sensitivity is strongly associated with actual error.

The contribution is a safety-layer concept rather than a new disease classifier. The same logic could be tested in imaging, genomics, drug response, or any setting where several credible model classes are available.

What the result does not mean

- It does not establish clinical safety or utility.
- It does not show that 20% is the best abstention rate.
- It does not prove that disagreement is calibrated under external distribution shift.
- It does not identify the biological cause of each hard case.
- It does not make repeated folds independent patients.
- It does not replace external validation or subgroup analysis.

These boundaries are part of the result, not optional disclaimers.

Robustness and auditability

The project is designed for audit. The raw file is checksummed; the protocol is immutable in the delivered batch; every prediction is saved; every plotted value has a CSV source; and numeric claims in the report are copied from a machine-readable metrics file.

The code uses deterministic seeds. Running the script recreates the held-out predictions, metrics, tables, and plots. Software-version differences may create tiny floating-point changes but should not alter the gate decision.

Limitations

1. **Single dataset:** the analysis lacks an independent institutional cohort.
2. **Curated features:** the inputs are engineered summaries, not raw clinical images.
3. **Small error count:** high accuracy produces a wide enrichment interval.
4. **No subgroup claims:** demographic variables are unavailable.
5. **No prospective workflow:** abstention costs and downstream review are not modeled.
6. **Benchmark age:** deployment conditions may differ from the source study.

External validation protocol

Freeze the present model and disagreement definition. Identify at least two independent cohorts with the same endpoint and compatible features. Apply the complete pipeline without tuning the 80th-percentile threshold on outcomes.

Required replication outputs

- Error enrichment and patient-clustered confidence interval.
- Coverage-risk curve with the 20% abstention point highlighted.
- Calibration and prevalence sensitivity.
- Performance by available clinical and technical subgroups.
- Failure audit for confidently wrong and highly disagreeing cases.

Ethics and use boundaries

The experiment uses public, de-identified data. No wet-lab work, patient contact, or intervention occurs. The output is research software and documentation.

Do not use this model for diagnosis, triage, reassurance, or treatment decisions. Clinical deployment would require representative external data, regulatory and institutional review, monitoring, security controls, and qualified medical oversight.

Reproduction guide

Run

```
cd projects/SP-001-model-agreement-instrument
python3 code/run_analysis.py
```

Key outputs

- `results/metrics.json`: gate metrics
- `data/processed/held_out_predictions.csv`: all held-out probabilities and disagreement scores
- `results/disagreement_deciles.csv`: plot data
- `results/coverage_risk.csv`: coverage-risk data
- `figures/`: publication figures

File manifest

The delivered directory separates protocol, code, raw data, processed data, results, figures, report, and paper. A SHA-256 manifest at the repository root permits integrity checking after transfer.

Directory	Purpose
protocol	Locked scientific plan
code	Exact reproducible analysis
data/raw	Original public inputs
data/processed	Held-out prediction dataset
results	Metrics and plot-support tables
figures	Publication graphics
report, paper	Human-readable deliverables

Future experiments

The result motivates harder tests that remain falsifiable:

1. Freeze the score and test cross-cohort transport.
2. Compare model disagreement with deep ensembles and conformal scores.
3. Measure whether disagreement adds information after single-model entropy.
4. Test subgroup-specific coverage-risk behavior.
5. Use disagreement clusters to identify feature-space blind spots.

Each extension should receive its own locked protocol. Failure must remain publishable internally rather than being tuned away.

Conclusion

Cross-model disagreement passed every preregistered criterion as an empirical uncertainty instrument in this benchmark. The top disagreement quintile had 18.08 times the error of remaining predictions, and excluding it reduced held-out error by 77.5% at roughly 80% coverage.

The result is genuinely useful because it transforms model diversity into a measurable warning signal. Its scientific value depends equally on the positive result and the stated limits: one curated dataset cannot establish clinical use. The next decisive step is unchanged-rule external validation.

Final status: SUCCESS

Reproducible proof of concept. External validation required.